Morgan Stanley | RESEARCH

IDEA

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Greater China Semiconductors | Asia Pacific

Cloud semis: time to turn constructive

With a potential cloud capex trough in 4Q/1Q26 and US tariff impact being priced in, we think it's time to turn positive from a top-down perspective. Rising inference demand to sustain CPU and non-standard AI server demand, on top of AI server rack ramp up. Upgrade Aspeed to OW and raise Montage PT.

Key Takeaways

- We think the market is underestimating CPU and non-standard AI server upside risk. The GB rack ramp up with potential longer life cycle is a clear positive.
- The higher content for performance/security is a longer-term tailwind for cloud.

Reversing our thesis on cloud semis: We took a more neutral view on cloud semi stocks in September 2024 (link), ahead of cloud capex and earnings revision peaks, and potential GB200 rack issue/inventory. But, we now expect the cycle to trough soon (likely 4Q), as earnings revisions look set to bottom and GB rack yield improves (see AI tracker). We think the stocks likely troughed after the US tariff announcement in April. This is supported by our recent US cloud capex tracker data, which indicated strong growth in 2025 and into 2026.

What about a potential 2H slow down? We have seen stronger-than-expected general server demand in 1H. We think this has been partly driven by inference demand (e.g. Deepseek), which is why we saw a surge in the new gen server mix from 1Q25. Alternative GPU solutions (e.g. RTX 6000 PRO) could also generate stronger momentum in 2H, not limited to China. The PCIE GPU server (2, 4 or 6 cards) should also sustain 2H demand. We thus think it's unlikely to see a significant decline in 2H demand.

Higher cloud semi content in longer term: We believe as servers enhance security functions (US brands and customers) this will result in higher semi content (e.g. PFR or AST2700). Companies with comprehensive product offerings and track records should gain share (e.g. Aspeed). We also expect China server brands will look to improve performance by adopting more advanced interfaces (e.g. MRDIMM). We think Montage will be a key beneficiary of this trend.

Upgrade Aspeed to OW (PT NT$5,000); Raise Montage PT to Rmb100: We think Aspeed's earnings downgrades have reached a bottom. We raise Aspeed's earnings by 9%/8%/31% in 2025/26/27e. Although the stock looks expensive at 47x our 2026e EPS vs. GC Semi fabless names at 26x, we move to OW as we think earnings momentum is the key for investor focus. We also raise Montage earnings by 9%/11% in 2026/27e. The stock trades on 38x our 2026e EPS, which looks attractive given our forecast for 25% earnings CAGR in 2024-27e.

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GREATER CHINA TECHNOLOGY SEMICONDUCTORS

Asia Pacific		
Industry View		In-Line

WHAT'S CHANGED

	From	To
Aspeed Technology (5274.TWO)		
Rating	Equal-weight	**Overweight**
Price Target	NT$3,000.00	**NT$5,000.00**
Montage Technology Co Ltd (688008.SS)	From	To
Price Target	Rmb88.00	**Rmb100.00**

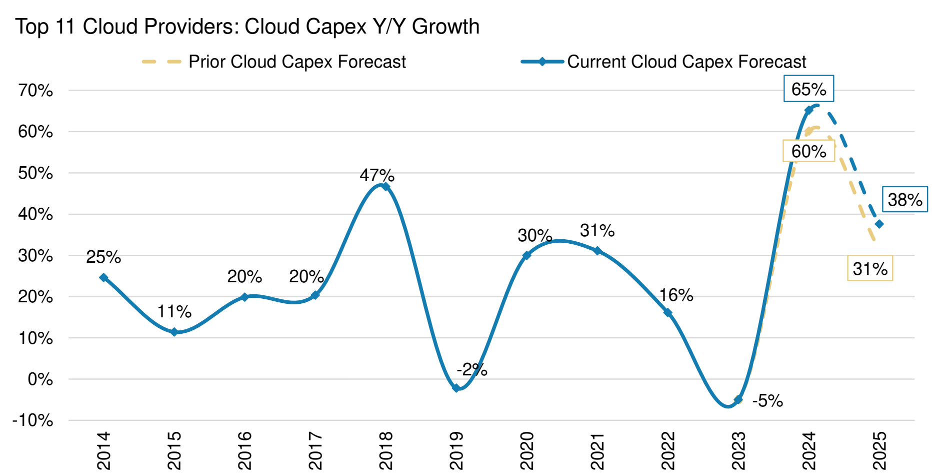
CIO Survey Results

US Cloud Capex Tracker

Our US Cloud Capex Tracker points to 38% Y/Y growth in CY25, up from +31% Y/Y a month ago (Exhibit 1), this is seven points, or US$29bn, higher than a month ago. For reference, this would make 2025 the third strongest year of capex growth in the past decade, trailing only 2024 (65% Y/Y) and 2018 (47% Y/Y). Furthermore, this implies that cumulatively, the top 11 global cloud players now expect to spend US$392bn on capex in 2025, or 16.6% of revenue (Exhibit 1), an all-time high, and nearly equivalent to total cloud capex in 2023 and 2024 combined.

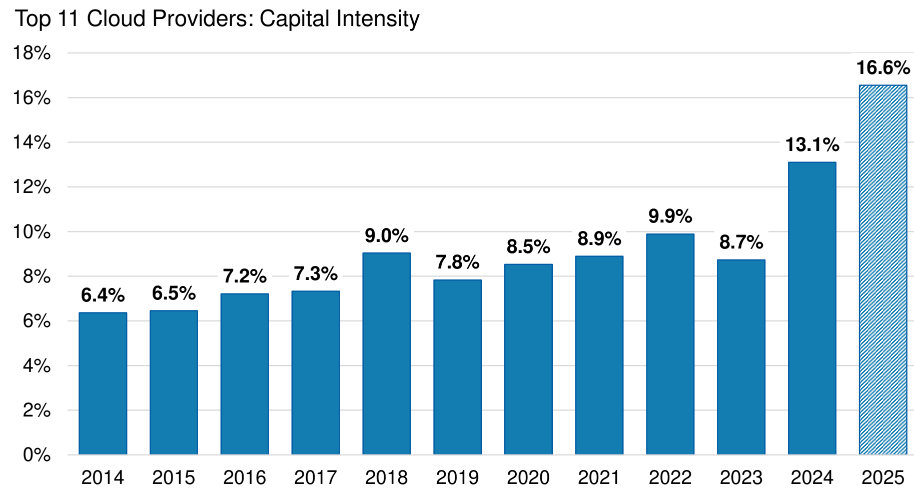
We track capex spending from the top 10 hyperscalers by server installed base. These include Google, Amazon, Microsoft, Meta Platforms, Tencent, Baidu, Apple, IBM, and Oracle. Dell Oro estimates that roughly ~75% of hyperscale capex goes towards data centers, which makes these companies a strong proxy for cloud data center equipment spending.

Exhibit 1: Post META/MSFT C1Q25 earnings, our cloud capex tracker now points to 38% Y/Y growth in 2025, up from the prior forecast of +31% Y/Y about a month ago.



Source: Company data, FactSet, Morgan Stanley Research estimates, irs.gov. Note: Cloud capex includes capex from Alphabet, Amazon, Microsoft, Meta Platforms, CoreWeave, Tencent, Baidu, Apple, IBM, and Oracle. Forward estimates include MSe for Tencent and Baidu, consensus est. for the others; Alphabet's CY25 capex is based on the company's guidance, while the remaining CY25 numbers are Consensus estimates.

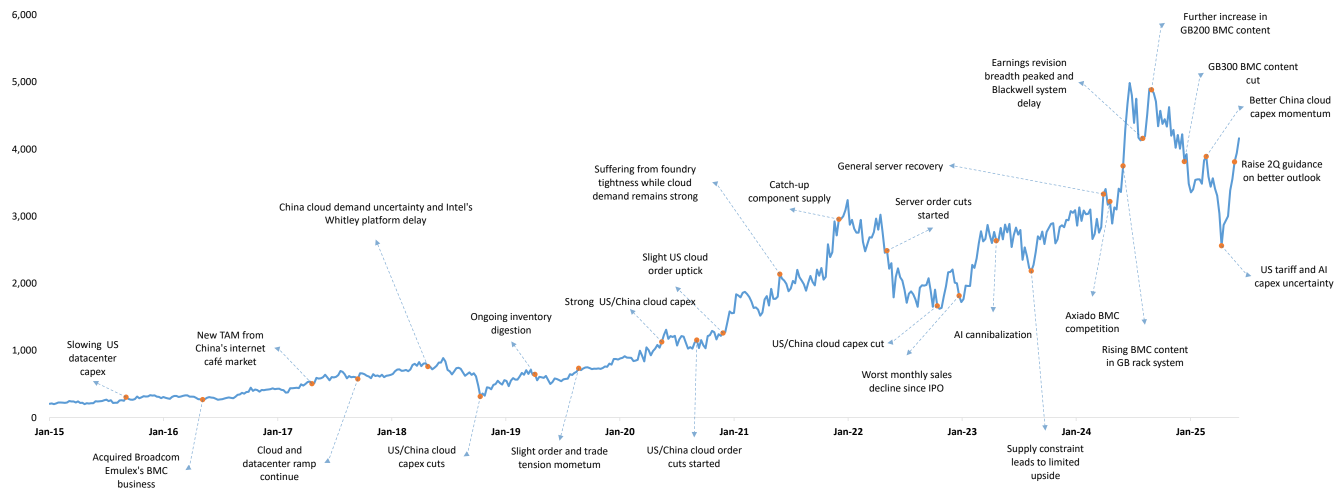
Exhibit 2: With stronger cloud capex now expected in 2025, overall capital intensity is now exceeding 16.6%, a new all-time high.



Source: FactSet, Morgan Stanley Research

Key Charts

Exhibit 3: Aspeed share price developments (NT$)



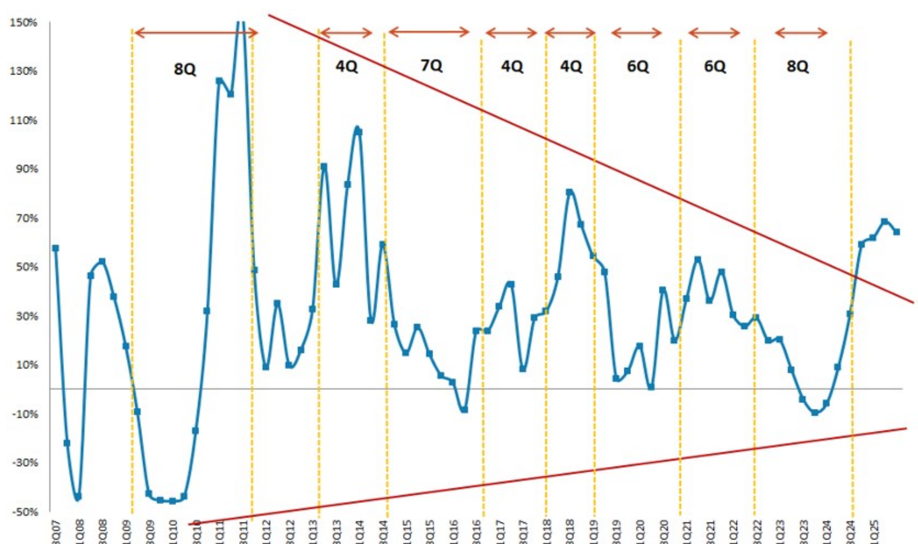
Source: Morgan Stanley Research

Exhibit 4: Aspeed share price downturn – 2018 vs. 2022 vs. 2024/25

	Peak to trough (current price) timing	Share price depreciation
2018 downturn	4 months	-65%
2022 downturn	6 months	-46%
2024/25 downturn	8 months	-43%

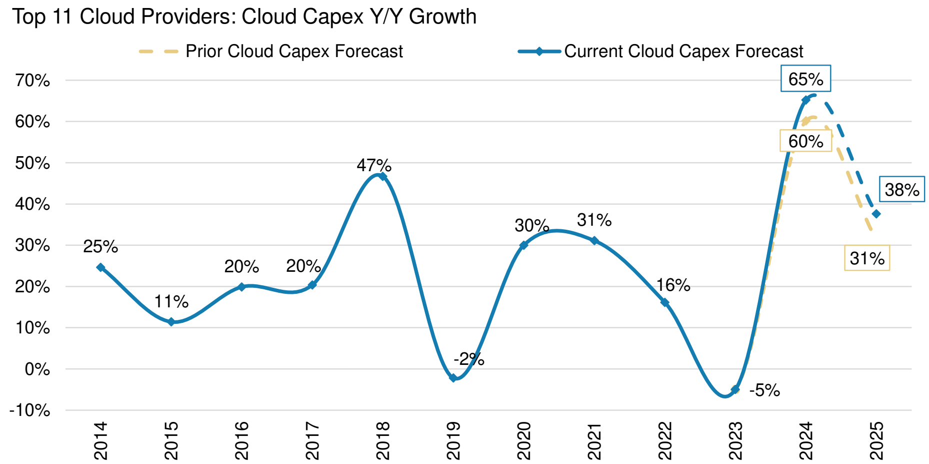
Source: Morgan Stanley Research

Exhibit 5: US cloud capex was +68% Y/Y in 4Q24, further decelerating to +64% Y/Y in 1Q25



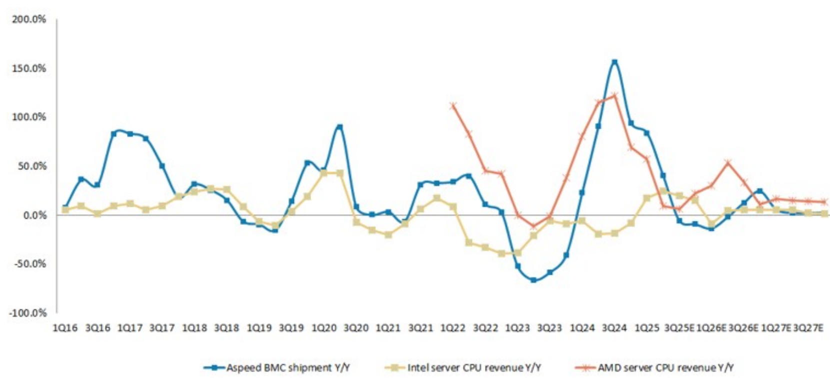
Source: Company data, Morgan Stanley Research; US cloud capex calculated from Alphabet, Microsoft, Amazon, and Meta.

Exhibit 6: After reflecting consensus estimates for Tencent, Alibaba, and Oracle capex, our cloud capex tracker now points to 65% Y/Y growth in 2024 (from 60% previously) and 38% Y/Y growth in 2025 (from 31% previously)



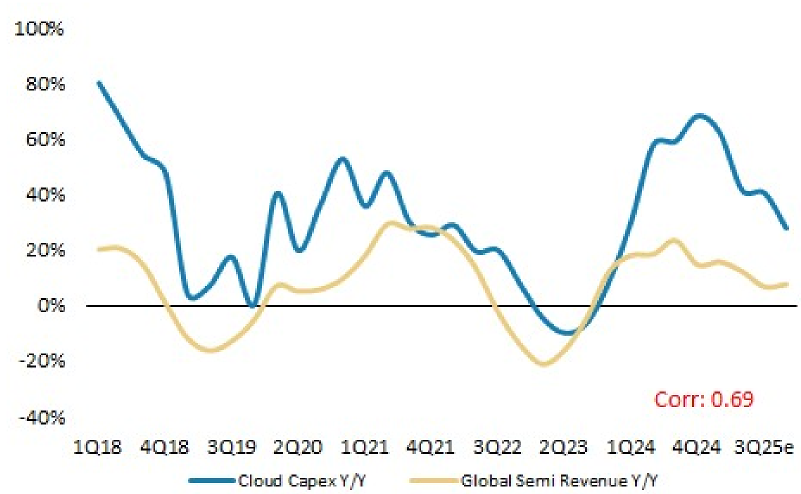
Source: Company data, FactSet, Morgan Stanley Research, irs.gov. Note: Cloud capex includes capex from Amazon, Alphabet, Microsoft, Meta Platforms, Tencent, Baidu, Apple, IBM, and Oracle. Forward estimates include MSe for Tencent and Baidu, consensus estimates for the others.

Exhibit 7: Quarterly Aspeed shipments Y/Y and Intel + AMD CPU revenue Y/Y



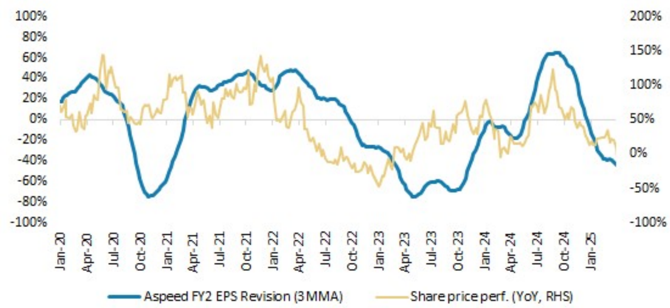
Source: Company data, Morgan Stanley Research. E=Morgan Stanley estimates.

Exhibit 8: Cloud capex has a major impact on global semi revenue, with the correlation being 0.69



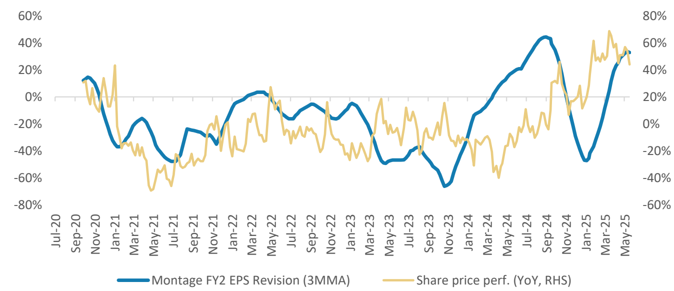
Source: Company data, Morgan Stanley Research. E=Morgan Stanley estimates.

Exhibit 9: Aspeed earnings estimate revision breadth vs. Y/Y share price performance



Source: FactSet, Morgan Stanley Research

Exhibit 10: Montage earnings estimate revision breadth vs. Y/Y share price performance



Source: FactSet, Morgan Stanley Research

Montage: Earnings Estimate Revisions

We raise our 2026 and 2027 EPS forecasts by 9% and 11%, respectively: With a potential cloud capex trough in 4Q/1Q26 and US tariff impact being priced in, we think this bodes well for the development and outlook for Montage. We also believe that China server brands are trying to improve performance by adopting more advanced interfaces (e.g. MRDIMM) in the long term and think Montage will be a key beneficiary of this trend.

Exhibit 11: Montage: Earnings Estimate Revisions

Rmb mn	Current 2025E	Previous 2025E	Diff.	Current 2026E	Previous 2026E	Diff.	Current 2027E	Previous 2027E	Diff.
Net sales	5,601	5,601	0%	7,082	6,724	5%	8,236	7,739	6%
COGS	2,205	2,205		2,830	2,697		3,303	3,092	
Gross profit	3,397	3,397	0%	4,252	4,027	6%	4,933	4,646	6%
Operating expenses	1,309	1,309		1,784	1,784		2,158	2,158	
Operating profit	2,088	2,088	0%	2,468	2,243	10%	2,774	2,488	12%
Non-op. income (exp.)	208	208		162	162		162	162	
Pretax Income	2,295	2,295	0%	2,630	2,405	9%	2,936	2,650	11%
Taxes	117	117		134	123		150	135	
Net income	2,200	2,200	0%	2,496	2,282	9%	2,787	2,515	11%
Reported EPS	**1.92**	**1.92**	**0%**	**2.18**	**1.99**	**9%**	**2.43**	**2.20**	**11%**
Margins									
Gross margin	60.6%	60.6%	0 ppt	60.0%	59.9%	0 ppt	59.9%	60.0%	0 ppt
Operating margin	37.3%	37.3%	0 ppt	34.8%	33.4%	1 ppt	33.7%	32.1%	2 ppt
Pretax margin	41.0%	41.0%	0 ppt	37.1%	35.8%	1 ppt	35.7%	34.2%	1 ppt
Net margin	39.3%	39.3%	0 ppt	35.2%	33.9%	1 ppt	33.8%	32.5%	1 ppt

Source: Morgan Stanley Research estimates

Exhibit 12: Montage Quarterly Financials

Rmb in million (YE Dec31)	1Q25	2Q25E	3Q25E	4Q25E	1Q26E	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025E	2026E	2027E
Total Revenues	1,222	1,325	1,472	1,581	1,559	1,686	1,873	1,964	1,776	1,978	2,182	2,300	3,639	5,601	7,082	8,236
Sequential Change	14.4%	8.4%	11.1%	7.4%	-1.4%	8.2%	11.1%	4.9%	-9.6%	11.4%	10.3%	5.4%				
Change vs Year Ago	65.8%	42.9%	62.5%	48.1%	27.5%	27.2%	27.2%	24.2%	13.9%	17.3%	16.5%	17.1%	59.2%	53.9%	26.4%	16.3%
Cost of Sales	483	523	577	621	626	675	747	782	712	791	873	926	1,524	2,205	2,830	3,303
Percent of Revenues	40%	39%	39%	39%	40%	40%	40%	40%	40%	40%	40%	40%	42%	39%	40%	40%
Gross Profit	739	803	895	960	933	1,011	1,127	1,182	1,064	1,187	1,309	1,373	2,115	3,397	4,252	4,933
Percent of Revenues	60.4%	60.6%	60.8%	60.7%	59.9%	59.9%	60.1%	60.2%	59.9%	60.0%	60.0%	59.7%	58.1%	60.6%	60.0%	59.9%
Incremental Margin	76%	62%	63%	59%	NM	61%	62%	61%	NM	61%	60%	55%	57%	65%	58%	59%
Total Opex	279	289	344	397	398	436	462	488	479	527	563	589	1,056	1,309	1,784	2,158
Percent of Revenues	22.8%	21.8%	23.4%	25.1%	25.5%	25.9%	24.7%	24.8%	27.0%	26.6%	25.8%	25.6%	29.0%	23.4%	25.2%	26.2%
R&D	153	193	233	273	263	293	313	333	313	353	383	403	763	852	1,202	1,452
Percent of Revenues	12.5%	14.6%	15.8%	17.3%	16.9%	17.4%	16.7%	17.0%	17.6%	17.8%	17.6%	17.5%	21.0%	15.2%	17.0%	17.6%
General & administrative	102	60	70	78	84	87	90	93	99	102	105	108	196	310	354	414
Percent of Revenues	8.3%	4.5%	4.8%	4.9%	5.4%	5.2%	4.8%	4.7%	5.6%	5.2%	4.8%	4.7%	5.4%	5.5%	5.0%	5.0%
Selling & marketing	24	36	41	46	51	56	59	62	67	72	75	78	96	147	228	292
Percent of Revenues	2.4%	2.3%	2.3%	2.3%	2.3%	2.3%	2.3%	2.3%	2.3%	2.3%	2.3%	2.3%	2.6%	2.6%	3.2%	3.5%
Operating Income	460	514	551	563	535	575	665	694	585	659	746	784	1,060	2,088	2,468	2,774
Percent of Revenues	37.6%	38.7%	37.4%	35.6%	34.3%	34.1%	35.5%	35.3%	32.9%	33.3%	34.2%	34.1%	29.1%	37.3%	34.8%	33.7%
Change vs Year Ago	144.6%	80.7%	98.0%	82.3%	16.3%	11.9%	20.6%	23.2%	9.3%	14.8%	12.3%	13.1%	164%	97%	18%	12%
Total Non-operating Income(Loss)	71	46	46	46	41	41	41	41	41	41	41	41	353	208	162	162
Profit Before Taxes	531	559	597	609	575	615	705	734	625	700	786	825	1,413	2,295	2,630	2,936
Percent of Revenues	43%	42%	41%	38%	37%	36%	38%	37%	35%	35%	36%	36%	39%	41%	37%	36%
Taxes	27	29	30	31	29	31	36	37	32	36	40	42	72	117	134	150
Tax Rate	5.1%	5.1%	5.1%	5.1%	5.1%	5.1%	5.1%	5.1%	5.1%	5.1%	5.1%	5.1%	5.1%	5.1%	5.1%	5.1%
Net Income, Cont Ops	504	531	566	578	546	584	669	697	593	664	746	783	1,341	2,178	2,496	2,787
Percent of Revenues	41%	40%	38%	37%	35%	35%	36%	35%	33%	34%	34%	34%	37%	39%	35%	34%
Reported Income	**525**	**531**	**566**	**578**	**546**	**584**	**669**	**697**	**593**	**664**	**746**	**783**	**1,412**	**2,200**	**2,496**	**2,787**
Percent of Revenues	43%	40%	38%	37%	35%	35%	36%	35%	33%	34%	34%	34%	39%	39%	35%	34%
Change vs Year Ago	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	213%	56%	13%	12%
Reported EPS	**0.46**	**0.46**	**0.49**	**0.50**	**0.48**	**0.51**	**0.58**	**0.61**	**0.52**	**0.58**	**0.65**	**0.68**	**1.24**	**1.92**	**2.18**	**2.43**
Change vs Year Ago	134%	43%	44%	31%	4%	10%	18%	21%	9%	14%	12%	12%	214%	54%	13%	12%

Source: Company data, Morgan Stanley Research (E) estimates

Valuation Methodology

We raise our Montage Technology price target to Rmb100 from Rmb88: The price target change reflects our revised earnings estimates, and we also raise the intermediate growth rate from 13.7% to 13.9% given our more optimistic view on cloud capex. Our price target is our base case value, derived from a residual income (RI) model as we believe an RI model best captures the company's sustainable growth over the longer term. We assume a 9% cost of equity (beta 1.3, risk-free rate 6.5%, and risk premium 2%), a payout ratio of 86%, and a terminal growth rate of 6%. These metrics are unchanged.

We also raise our bull case value to Rmb125 from Rmb100, and our bear case value reduces to Rmb45 from Rmb40. These changes are in line with changes to our base case value.

Exhibit 13: Montage RI Valuation

Rmb mn	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Total Equity	**13,253**	**15,406**	**17,850**	**18,568**	**19,385**	**20,316**	**21,375**	**22,583**	**23,958**	**25,524**	**27,308**	**29,340**
Net Profit	**2,200**	**2,496**	**2,787**	**3,174**	**3,615**	**4,118**	**4,690**	**5,342**	**6,085**	**6,930**	**7,894**	**8,991**
ROAE	**17.8%**	**17.4%**	**16.8%**	**17.4%**	**19.1%**	**20.7%**	**22.5%**	**24.3%**	**26.1%**	**28.0%**	**29.9%**	**31.7%**
Residual Income	1,008	1,116	1,195	1,505	1,866	2,277	2,743	3,272	3,872	4,555	5,330	6,211
Spread	**8.8%**	**8.4%**	**7.8%**	**8.4%**	**10.1%**	**11.7%**	**13.5%**	**15.3%**	**17.1%**	**19.0%**	**20.9%**	**22.7%**
Ending Equity Capital	13,253											
PV of Forecast Period	15,711											
PV of Continuing Value	85,044											
Equity Value	**114,007**											
No. of Shares	1,145											
Projected Price	**100**											

Source: Company data, Morgan Stanley Research estimates

Risk Reward – Montage Technology Co Ltd (688008.SS)

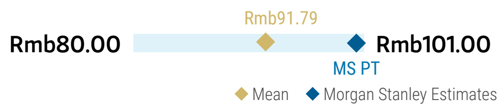
Tapping into PCIE Switch Market; OW

PRICE TARGET Rmb100.00

Base case, residual income model. Key assumptions:
- 9% cost of equity (beta 1.3, risk-free rate 6.5%, and risk premium 2%)
- Payout ratio of 86%
- Medium-term growth rate of 13.9%
- Terminal growth rate of 6%

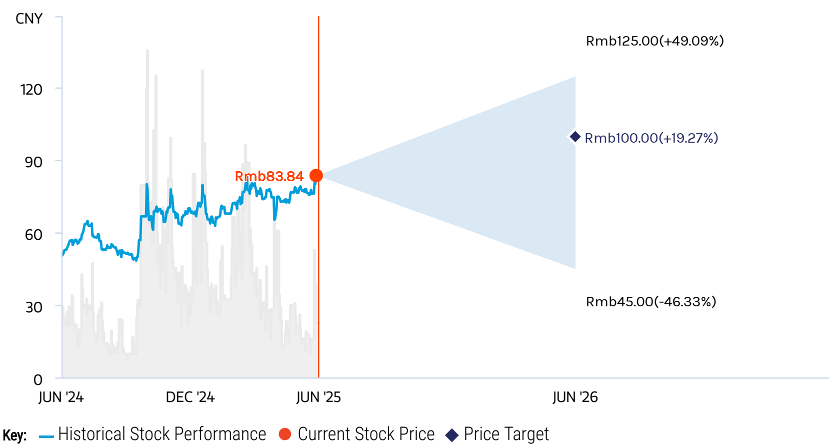
We believe these assumptions are justified, given cloud capex growth, DRAM interface technology migration, and Montage's strong position in China's data center semi localization.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART

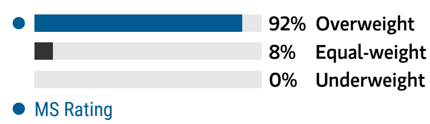


Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We view Montage as a proxy for DDR5 chips in Greater China.
- It has launched its third-generation DDR5 RCD chip and expects volume to surpass Gen 2 in 2H25. An increase in the mix of Gen 3 chips will help margins.
- Montage should enjoy interface spec migration in light of its DDR5 Gen 3 interface chips and CXL.
- New products like MRCD, MDB, PCIe Gen5 and Gen6 retimers, and PCIe switches offer new growth opportunities.
- With a potential cloud capex trough in 4Q/1Q26 and US tariff impact being priced in, we think this bodes well for the development and outlook for Montage.
- Valuation looks attractive at 44x 2025e EPS, >1 SD below its historical average of 74x since 2020.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE Rmb125.00

65x 2025e EPS

We expect a 60%+ CAGR for interface chips revenue in 2024-27e: DDR5 penetration ramp-up is faster than expected. Large-scale adoption of its PCIe retimer by overseas customers.

BASE CASE Rmb100.00

52x 2025e EPS

We expect 33% CAGR for interface chips revenue in 2024-27e: Montage is likely to benefit from fast PC DDR5 migration. Interface chips gross margin at around 62% in 2024-27e. New products contribute meaningfully to revenue in 2025.

BEAR CASE Rmb45.00

23x 2025e EPS

We expect a revenue CAGR below 15% for interface chips in 2024-27e: We expect peers to become more aggressive on pricing and thus gross margin to trend down. Limited contribution from new products.

Risk Reward – Montage Technology Co Ltd (688008.SS)

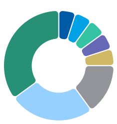
KEY EARNINGS INPUTS

Drivers	2024	2025e	2026e	2027e
Register clock driver shipment (Rmb, mn)	163	231	256	238
Data buffer shipment (Rmb, mn)	53	75	90	90

INVESTMENT DRIVERS

- Cloud capex strength
- DRAM technology migration
- CPU technology migration

GLOBAL REVENUE EXPOSURE



- 0-10% Europe ex UK
- 0-10% India
- 0-10% Japan
- 0-10% MEA
- 0-10% UK
- 10-20% North America
- 20-30% APAC, ex Japan, Mainland China and India
- 30-40% Mainland China

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

4/5 MOST | 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Faster-than-expected phase-out of US peers
- Faster-than-expected spec migration

RISKS TO DOWNSIDE

- Weaker-than-expected cloud demand
- Slower-than-expected DRAM interface technology migration
- Intensifying competition from US peers

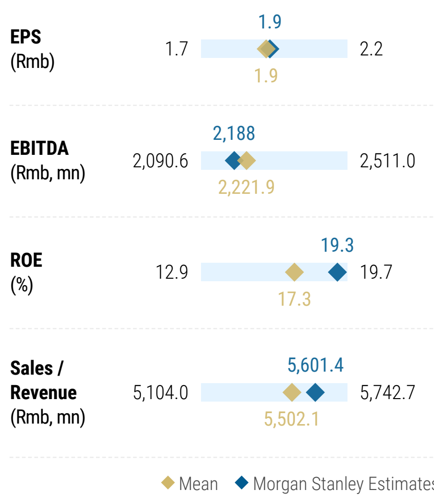
OWNERSHIP POSITIONING

Inst. Owners, % Active 98.1%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2025e



Source: Refinitiv, Morgan Stanley Research

Montage: Financial Summary

Income Statement

Rmb mn (Years End Dec)	2024	2025E	2026E	2027E
Net sales	3,639	5,601	7,082	8,236
COGS	(1,524)	(2,205)	(2,830)	(3,303)
Gross profit	2,115	3,397	4,252	4,933
Operating expenses	(1,056)	(1,309)	(1,784)	(2,158)
Operating income	**1,060**	**2,088**	**2,468**	**2,774**
Non-operating income	353	208	162	162
Pre-tax income	1,413	2,295	2,630	2,936
Income tax	72	117	134	150
Net Income	**1,341**	**2,178**	**2,496**	**2,787**
Minority Interest	(71)	(22)	0	0
Reported net Income	**1,270**	**2,157**	**2,496**	**2,787**
Adj.wtd.avg.shrs(m)	1,134	1,145	1,145	1,145
Reported EPS (NT$)	**1.24**	**1.92**	**2.18**	**2.43**

Balance Sheet

Rmb mn (Years End Dec)	2024	2025E	2026E	2027E
Cash	5,723	7,207	9,101	11,341
Mkt Securities	1,677	1,677	1,677	1,677
AR/NR	388	525	664	772
Inventory	352	604	775	905
Other	1,321	1,321	1,321	1,321
Current Assets	**9,461**	**11,334**	**13,538**	**16,015**
Long-term investments	109	109	109	109
Fixed assets	582	602	622	642
Deferred assets	83	83	83	83
Other assets	1,983	1,983	1,983	1,983
Total Assets	**12,219**	**14,112**	**16,335**	**18,833**
S/T borrowings	0	0	0	0
AP/NP	211	248	318	371
Other ST liabilities	469	469	469	469
LT debt	0	0	0	0
Other LT liabilities	143	143	143	143
Common shares	1,145	1,145	1,145	1,145
Total Liabilities	**822**	**859**	**929**	**982**
Additional capital	5,626	5,626	5,626	5,626
Retained earning	4,518	6,375	8,528	10,972
Other shareholders' equity	107	107	107	107
Total Equity	**11,397**	**13,253**	**15,406**	**17,850**
Total Liab. & Shrhldr's Equity	**12,219**	**14,112**	**16,335**	**18,833**

E = Morgan Stanley Research Estimates

Source: Morgan Stanley Research, Company Data

Cash Flow Statement

Rmb mn (Years End Dec)	2024	2025E	2026E	2027E
Cashflow from Operations	**1,691**	**1,948**	**2,366**	**2,723**
Net profits	1,412	2,200	2,496	2,787
Depreciation	67	72	77	82
Working Capital Change	(923)	(352)	(240)	(184)
Other adjustments	1,135	28	33	38
Cashflow from Investing	**(443)**	**(120)**	**(130)**	**(140)**
Capex	(379)	(120)	(130)	(140)
Change of LT Investment	(39)	0	0	0
Change of ST Investment	(26)	0	0	0
Other adjustments	1	0	0	0
Cashflow from financing	**(277)**	**(343)**	**(342)**	**(342)**
Increase in L/T debt	0	0	0	0
Increase in S/T debt	0	0	0	0
Cash Dividend Paid	0	(343)	(342)	(342)
Issuance of stock	194	0	0	0
Other adjustments	(471)	0	0	0
Exchange rate adjustment	62	0	0	0
Net change in cash	**1,033**	**1,484**	**1,893**	**2,240**

Financial Ratios

	2024	2025E	2026E	2027E
Growth(%)				
Turnover	59.2	53.9	26.4	16.3
Operating profits	163.8	97.0	18.2	12.4
EPS	214.0	54.4	13.5	11.7
Margins (%)				
Gross Margin	58.1	60.6	60.0	59.9
Operating Margin	29.1	37.3	34.8	33.7
Pretax Margin	38.8	41.0	37.1	35.7
Net Margin	34.9	38.5	35.2	33.8
Return (%)				
ROAE	11.8	17.5	17.4	16.8
ROAA	11.1	16.4	16.4	15.8
Gearing (%)				
Net Debt/Equity	(50.2)	(54.4)	(59.1)	(63.5)
Liabilities/Equity	7.2	6.5	6.0	5.5
Ratios (X)				
Current ratio	13.9	15.8	17.2	19.1
Quick ratio	9.0	10.8	12.4	14.4
Others				
AR/NR Turnover (days)	34	34	34	34
Inventory Turnover (days)	100	100	100	100
AP Turnover (days)	41	41	41	41
Cash Conversion (days)	93	93	93	93

Aspeed: Earnings Estimate Revisions

We raise our 2025, 2026 and 2027 EPS estimates by 9%, 8% and 31%, respectively: Besides ongoing strength in general servers into 2H, alternative GPU solutions (e.g. RTX 6000 PRO) could also generate stronger momentum in 2H, not limited to China. PCIE GPU servers (2, 4 or 6 cards) should also see sustained 2H demand. We thus think it's unlikely to see a significant decline in demand and revenue in 2H. In the long run, we believe enhancing server security functions (US brands and customers) will result in higher semi content (e.g. PFR or AST2700). Given Aspeed's comprehensive product offering and track record we think it will gain share and therefore upgrade the stock to Overweight from Equal-weight.

Exhibit 14: Aspeed: Earnings estimate revisions

	Current	Previous		Current	Previous		Current	Previous	
NT$ mn	**2025E**	**2025E**	**Diff.**	**2026E**	**2026E**	**Diff.**	**2027E**	**2027E**	**Diff.**
Net sales	8,045	7,345	10%	9,769	8,988	9%	12,107	9,735	24%
COGS	2,701	2,530		3,265	3,033		3,990	3,222	
Gross profit	5,344	4,815	11%	6,504	5,954	9%	8,117	6,513	25%
Operating expenses	1,373	1,160		1,455	1,280		1,683	1,623	
Operating profit	3,971	3,656	9%	5,049	4,674	8%	6,434	4,889	32%
Non-op. income (exp.)	155	102		142	142		62	62	
Pretax Income	4,125	3,758	10%	5,191	4,816	8%	6,497	4,951	31%
Taxes	777	695		939	872		1,176	896	
Net income	3,348	3,062	9%	4,252	3,944	8%	5,321	4,055	31%
Reported EPS	**88.60**	**81.04**	**9%**	**112.50**	**104.38**	**8%**	**140.79**	**107.30**	**31%**
Margins									
Gross margin	66.4%	65.6%	1 ppt	66.6%	66.2%	0 ppt	67.0%	66.9%	0 ppt
Operating margin	49.4%	49.8%	0 ppt	51.7%	52.0%	0 ppt	53.1%	50.2%	3 ppt
Pretax margin	51.3%	51.2%	0 ppt	53.1%	53.6%	0 ppt	53.7%	50.9%	3 ppt
Net margin	41.6%	41.7%	0 ppt	43.5%	43.9%	0 ppt	43.9%	41.7%	2 ppt

Source: Company data, Morgan Stanley Research (E) estimates

Exhibit 15: Aspeed: Quarterly financials

NT$ in million (YE DEC31)	1Q25E	2Q25E	3Q25E	4Q25E	1Q26E	2Q26E	3Q25E	4Q26E	2024	2025E	2026E	2027E
Total Revenues	2,065	2,029	2,021	1,930	1,988	2,294	2,615	2,872	6,460	8,045	9,769	12,107
Sequential Change	-1.8%	-1.8%	-0.4%	-4.5%	3.0%	15.4%	14.0%	9.8%				
Change vs Year Ago	103.8%	49.6%	1.8%	-8.3%	-3.7%	13.1%	29.4%	48.9%	106.4%	24.5%	21.4%	23.9%
Cost of Sales	697	683	677	643	672	769	872	952	2,306	2,701	3,265	3,990
Percent of Revenues	34%	34%	33%	33%	34%	34%	33%	33%	36%	34%	33%	33%
Gross Margin	1,368	1,345	1,344	1,287	1,317	1,524	1,743	1,921	4,154	5,344	6,504	8,117
Percent of Revenues	66.2%	66.3%	66.5%	66.7%	66.2%	66.5%	66.7%	66.9%	64.3%	66.4%	66.6%	67.0%
Incremental Margin	NM	NM	NM	NM	51%	68%	68%	69%	64%	75%	67%	69%
Total Opex	341	339	349	345	350	356	372	377	1,235	1,373	1,455	1,683
Percent of Revenues	16.5%	16.7%	17.3%	17.9%	17.6%	15.5%	14.2%	13.1%	19.1%	17.1%	14.9%	13.9%
R&D	180	180	190	190	230	230	240	240	829	740	940	940
Percent of Revenues	8.7%	8.9%	9.4%	9.8%	11.6%	10.0%	9.2%	8.4%	12.8%	9.2%	9.6%	7.8%
General & administrative	70	70	70	70	80	80	80	80	286	280	320	440
Percent of Revenues	3.4%	3.5%	3.5%	3.6%	4.0%	3.5%	3.1%	2.8%	4.4%	3.5%	3.3%	3.6%
Selling & marketing	91	89	89	85	40	46	52	57	120	353	195	303
Percent of Revenues	4.4%	4.4%	4.4%	4.4%	2.0%	2.0%	2.0%	2.0%	1.9%	4.4%	2.0%	2.5%
Operating Income	1,027	1,006	996	942	967	1,168	1,371	1,543	2,918	3,971	5,049	6,434
Percent of Revenues	49.7%	49.6%	49.3%	48.8%	48.6%	50.9%	52.4%	53.7%	45.2%	49.4%	51.7%	53.1%
Change vs Year Ago	165.8%	73.5%	4.6%	-5.9%	-5.9%	16.1%	37.7%	63.9%	170%	36%	27%	27%
Total Non-operating Income(Loss)	78	26	26	26	36	36	36	36	249	155	142	62
Profit Before Taxes	1,105	1,032	1,021	967	1,002	1,204	1,406	1,579	3,167	4,125	5,191	6,497
Percent of Revenues	54%	51%	51%	50%	50%	52%	54%	55%	49%	51%	53%	54%
Taxes	218	191	189	179	185	217	253	284	596	777	939	1,176
Tax Rate	19.8%	18.5%	18.5%	18.5%	18.5%	18.0%	18.0%	18.0%	18.8%	18.8%	18.1%	18.1%
Net Income, Cont Ops	887	841	832	788	817	987	1,153	1,295	2,571	3,348	4,252	5,321
Percent of Revenues	43%	41%	41%	41%	41%	43%	44%	45%	40%	42%	44%	44%
Reported Income (TW GAAP)	**887**	**841**	**832**	**788**	**817**	**987**	**1,153**	**1,295**	**2,571**	**3,348**	**4,252**	**5,321**
Percent of Revenues	43%	41%	41%	41%	41%	43%	44%	45%	40%	42%	44%	44%
Change vs Year Ago	0%	0%	0%	0%	0%	0%	0%	0%	155%	30%	27%	25%
Reported EPS (NT$, TW GAAP)	**23.47**	**22.25**	**22.02**	**20.86**	**21.62**	**26.12**	**30.51**	**34.26**	**68.05**	**88.60**	**112.50**	**140.79**
Change vs Year Ago	127%	66%	14%	-16%	-8%	17%	39%	64%	155%	30%	27%	25%

Source: Company data, Morgan Stanley Research (E) estimates

Aspeed: Valuation Methodology

We increase our price target from NT$3,000 to NT$5,000: This factors in changes to our 2025-27e estimates, while we also increase the intermediate growth rate from 14.9% to 17.2% and payout ratio from 64% to 80%. The new price target implies 56x 2025e EPS, which is above the 49x historical average of 1-year forward P/E since 2020 but we think is justifiable in view of improving demand. Other assumptions are unchanged, including:

- Cost of equity of 9.8% (2% risk-free rate, 6% risk premium, 1.3 beta).
- Terminal growth rate of 5.2%.

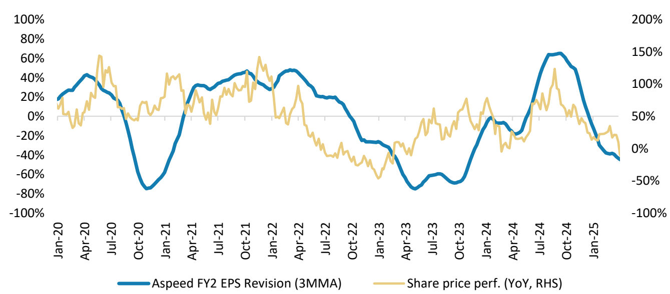
We also raise our bull and bear case values from NT$4,220 and NT$1,330, respectively, to NT$6,000 and NT$2,000, implying 68x and 23x 2025e EPS.

Exhibit 16: Aspeed: Residual income model

NT$million	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Total Equity	**6,841**	**8,298**	**10,069**	**11,818**	**13,868**	**16,271**	**19,087**	**22,388**	**26,257**	**30,792**	**36,108**	**42,338**
Net Profit	**3,348**	**4,252**	**5,321**	**6,236**	**7,310**	**8,567**	**10,042**	**11,770**	**13,796**	**16,170**	**18,953**	**22,215**
ROAE	**53.7%**	**56.2%**	**57.9%**	**57.0%**	**56.9%**	**56.9%**	**56.8%**	**56.8%**	**56.7%**	**56.7%**	**56.7%**	**56.6%**
Residual Income	2,475	3,174	3,996	4,753	5,571	6,529	7,651	8,967	10,510	12,318	14,437	16,920
Spread	**43.9%**	**46.4%**	**48.2%**	**47.2%**	**47.1%**	**47.1%**	**47.0%**	**47.0%**	**46.9%**	**46.9%**	**46.9%**	**46.9%**
Ending Equity Capital	6,841											
PV of Forecast Period	42,754											
PV of Continuing Value	139,353											
Equity Value	**188,948**											
No. of Shares	38											
Projected Price	**5,000**											

Source: Company data, Morgan Stanley Research estimates

Exhibit 17: Aspeed: Earnings revision breadth



Source: Company data, Morgan Stanley Research estimates

Exhibit 18: Aspeed: P/E ratio

Source: Company data, Morgan Stanley Research estimates

Risk Reward – Aspeed Technology (5274.TWO)

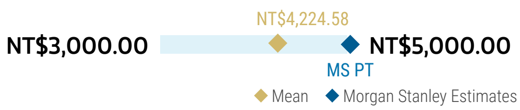
Time to turn constructive; OW

PRICE TARGET NT$5,000.00

Base case, residual income model. Key assumptions:

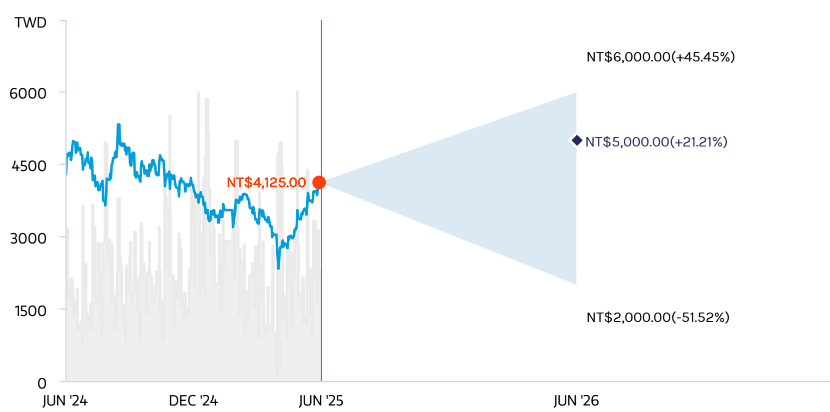
- Cost of equity of 9.8% (2.0% risk-free rate, 6% risk premium, 1.3 beta)
- Medium-term growth rate of 17.2%
- Terminal growth rate of 5.2%
- Cash payout ratio of 80%

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



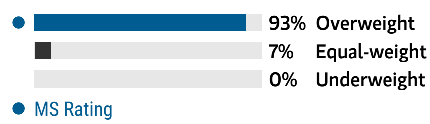
Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We view Aspeed as a distinctive NVIDIA-related play, being the major BMC supplier for AI server and a key supplier to help realize Omniverse adoption by leveraging its Coupola360 products.
- With a potential cloud capex trough in 4Q/1Q26 and US tariff impact priced in, we think it's time to turn positive from a top-down perspective. Rising inference demand could sustain CPU and non-standard AI server demand, on top of AI server rack ramp up.
- We believe enhancing server security functions (US brands and customers) will result in higher semi content.
- Our price target implies 56x 2025e EPS, above the 49x historical avg 1-year fwd P/E since 2020 but we think is justified given improving demand.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$6,000.00

68x 2025e EPS

Booming data center demand with significant share gains; meaningful OPM expansion; new businesses take off faster than expected: Significant earnings growth from data center demand and market share gains in white brands and US server brands. Margins improve considerably. New business growth accelerates. We also assume AI servers fully ramp up and clients execute new server platform migration faster than expected.

BASE CASE NT$5,000.00

56x 2025e EPS

Cloud/traditional servers bottomed out after a year of correction with momentum resuming; flattish OPM; continues to be beneficiary of GB200; new business bears fruit: 1) Revenue growth of 25% Y/Y in 2025 after strong growth in 2024; 2) gross margin of 66% in 2025 vs. 64% in 2024; 3) GB200 to see more demand on BMC.

BEAR CASE NT$2,000.00

23x 2025e EPS

Disappointing end-demand, with share loss; operating margin pressure; new businesses face challenges: Revenue growth decelerates amid slow data center demand, and share loss amid a weakening macro environment. OPM pressure emerges amid intense competition. New businesses stay flattish as end-markets take off more slowly.

Risk Reward – Aspeed Technology (5274.TWO)

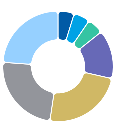
KEY EARNINGS INPUTS

Drivers	2024	2025e	2026e	2027e
BMC revenue (NT$, mn)	5,244	6,927	8,373	10,354
360 degree SoC revenue (NT$, mn)	367	352	484	643
PC/AV extension IC revenue (NT$, mn)	147	112	116	110

INVESTMENT DRIVERS

- Expanding share in baseboard management controller (BMC) market
- Cloud spending acceleration from customers
- Margin expansion
- New business progress

GLOBAL REVENUE EXPOSURE



- 0-10% Latin America
- 0-10% MEA
- 0-10% UK
- 10-20% Europe ex UK
- 20-30% APAC, ex Japan, Mainland China and India
- 20-30% Mainland China
- 20-30% North America

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

3/5 MOST	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger cloud demand
- Faster-than-expected spec migration
- Mild competition

RISKS TO DOWNSIDE

- Softening cloud demand
- Slower-than-expected spec migration
- Intensified competition
- Further policy tightening in China

OWNERSHIP POSITIONING

Inst. Owners, % Active 80.8%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2025e

	Low	Morgan Stanley Estimates	Mean	High
Sales / Revenue (NT$, mn)	7,005	8,045	7,900	8,422
EBITDA (NT$, mn)	3,218	3,971	3,927	4,612
Net income (NT$, mn)	2,678	3,348	3,196	3,438
EPS (NT$)	70.81	88.60	84.52	90.91

Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Aspeed: Financial Summary

Income Statement

NT$mn (Years End Dec)	2023	2024	2025E	2026E	2027E
Net sales	3,130	6,460	8,045	9,769	12,107
COGS	(1,122)	(2,306)	(2,701)	(3,265)	(3,990)
Gross profit	2,008	4,154	5,344	6,504	8,117
Operating expenses	(928)	(1,235)	(1,373)	(1,455)	(1,683)
Operating income	**1,080**	**2,918**	**3,971**	**5,049**	**6,434**
Non-operating income	115	249	155	142	62
Pre-tax income	1,195	3,167	4,125	5,191	6,497
Income tax	188	596	777	939	1,176
Reported net Income	**1,007**	**2,571**	**3,348**	**4,252**	**5,321**
Adj.wtd.avg.shrs(m)	38	38	38	38	38
Reported EPS (NT$)	**26.66**	**68.05**	**88.60**	**112.50**	**140.79**

Balance Sheet

NT$mn (Years End Dec)	2023	2024	2025E	2026E	2027E
Cash	1,612	3,659	4,862	6,001	7,354
Mkt Securities	373	352	352	352	352
AR/NR	716	1,437	1,341	1,628	2,018
Inventory	306	357	388	469	574
Other	417	181	181	181	181
Current Assets	**3,423**	**5,987**	**7,125**	**8,633**	**10,479**
Long-term investments	0	0	0	0	0
Fixed assets	198	355	385	415	445
Deferred assets	42	69	69	69	69
Other assets	881	1,316	1,316	1,316	1,316
Total Assets	**4,544**	**7,726**	**8,893**	**10,432**	**12,308**
S/T borrowings	0	0	0	0	0
AP/NP	242	426	391	473	578
Other ST liabilities	438	1,457	1,457	1,457	1,457
LT debt	0	0	0	0	0
Other LT liabilities	31	204	204	204	204
Common shares	378	378	378	378	378
Total Liabilities	**710**	**2,087**	**2,052**	**2,134**	**2,239**
Additional capital	2,340	2,440	2,440	2,440	2,440
Retained earning	1,075	2,851	4,053	5,510	7,281
Other shareholders' equity	42	(31)	(31)	(31)	(31)
Total Equity	**3,834**	**5,639**	**6,841**	**8,298**	**10,069**
Total Liab. & Shrhldr's Equity	**4,544**	**7,726**	**8,893**	**10,432**	**12,308**

E = Morgan Stanley Research Estimates

Source: Morgan Stanley Research, Company Data

Cash Flow Statement

NT$mn (Years End Dec)	2023	2024	2025E	2026E	2027E
Cashflow from Operations	**519**	**3,145**	**3,379**	**3,965**	**4,932**
Net profits	1,007	2,571	3,348	4,252	5,321
Depreciation	128	121	0	0	0
Working Capital Change	392	(66)	30	(287)	(389)
Other adjustments	(1,007)	519	0	0	0
Cashflow from Investing	**(0)**	**(345)**	**(30)**	**(30)**	**(30)**
Capex	(83)	(25)	(30)	(30)	(30)
Change of LT Investment	0	0	0	0	0
Change of ST Investment	177	43	0	0	0
Other adjustments	(95)	(363)	0	0	0
Cashflow from financing	**(1,728)**	**(781)**	**(2,147)**	**(2,795)**	**(3,549)**
Increase in L/T debt	0	0	0	0	0
Increase in S/T debt	0	0	0	0	0
Cash Dividend Paid	(1,702)	(756)	(2,147)	(2,795)	(3,549)
Dir& Emp Bonus Paid	0	0	0	0	0
Issuance of stock	0	0	0	0	0
Other adjustments	(26)	(25)	0	0	0
Exchange rate adjustment	(8)	30	0	0	0
Net change in cash	**-1,217**	**2,048**	**1,202**	**1,140**	**1,353**

Financial Ratios

	2023	2024	2025E	2026E	2027E
Growth(%)					
Turnover	-39.9	106.4	24.5	21.4	23.9
Operating profits	-55.9	170.3	36.1	27.1	27.4
EPS	-52.2	155.2	30.2	27.0	25.1
Margins (%)					
Gross Margin	64.1	64.3	66.4	66.6	67.0
Operating Margin	34.5	45.2	49.4	51.7	53.1
Pretax Margin	38.2	49.0	51.3	53.1	53.7
Net Margin	32.2	39.8	41.6	43.5	43.9
Return (%)					
ROAE	24.2	54.3	53.7	56.2	57.9
ROAA	19.2	41.9	40.3	44.0	46.8
Gearing (%)					
Net Debt/Equity	(42.0)	(64.9)	(71.1)	(72.3)	(73.0)
Liabilities/Equity	18.5	37.0	30.0	25.7	22.2
Ratios (X)					
Current ratio	5.0	3.2	3.9	4.5	5.2
Quick ratio	3.4	2.7	3.4	4.0	4.6
Others					
AR/NR Turnover (days)	99	61	61	61	61
Inventory Turnover (days)	115	52	52	52	52
AP Turnover (days)	91	53	53	53	53
Cash Conversion (days)	123	60	60	60	60

Valuation Methodology and Risks

Apple, Inc. (AAPL.O)

Our $235 PT is based a 7.9x EV/Sales FY26 multiple, which is derived from a regression of tech and consumer platform peers. Our price target implies 29.2x P/E on $8.06 CY26 EPS.

Risks to Upside

- iPhone 17 outperforms expectations
- Apple Intelligence adoption surprises to the upside
- Apple pulls forward form factor changes
- Services growth re-accelerates despite tougher compares
- Gross margins surprise positively

Risks to Downside

- Weak consumer spending limits iPhone upgrade rates
- Limited progress on AI features
- Geopolitical tensions/tariffs
- Increased regulation, particularly with GOOGL TAC and App Store

CoreWeave (CRWV.O)

24x EV/CY35 FCF of $5.8B (implies 15x EV/CY26 EBIT, a discount to peers)

Risks to Upside

- Monetizable software
- Differentiation in inference
- Go-to-market progress and lower customer concentration
- Evidence of GPU monetization over six years

Risks to Downside

- Normalization in AI demand
- Trends of shortening useful life of AI servers in the industry
- Commentary from large customers around intentions to in-house AI compute in the future
- Difficulty in raising capital at an attractive or improving cost of debt

IBM (IBM.N)

Our $233 price target reflects a 15.0x P/FCF 2026 multiple of $15.55 FCF/share.

Risks to Upside

- Strength in Gen AI adds incremental growth to SW and Consulting
- Consulting growth surprises positively
- Incremental M&A accelerates growth

- IBM upsides on FCF

Risks to Downside

- Macro disruption leads to IT spending cuts
- Accelerated cloud cannibalization of mainframe & legacy software
- Consulting declines throughout the year
- AI cannibalizes the existing Consulting businesses

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

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Global Stock Ratings Distribution

(as of May 31, 2025)

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For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1493	40%	379	46%	25%	698	41%
Equal-weight/Hold	1650	44%	372	45%	23%	782	46%
Not-Rated/Hold	4	0%	0	0%	0%	2	0%
Underweight/Sell	602	16%	74	9%	12%	235	14%
Total	3,749		825			1717	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

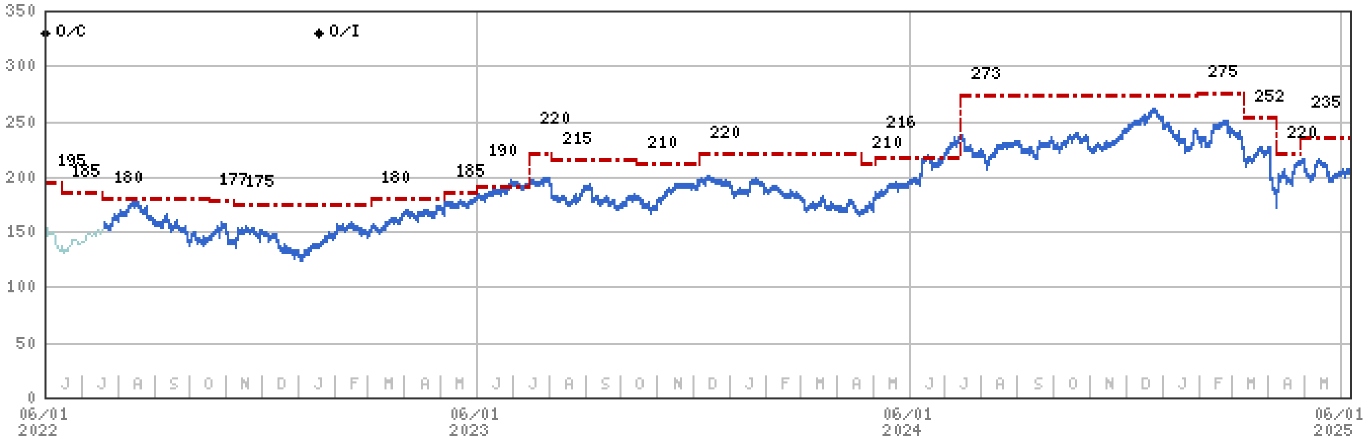
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Apple, Inc. (AAPL.O) - As of 06/09/25 GMT in USD
Industry : IT Hardware



Stock Rating History: 6/1/20 : O/I; 10/12/20 : O/A; 6/10/21 : O/I; 10/5/21 : O/C; 1/18/23 : O/I

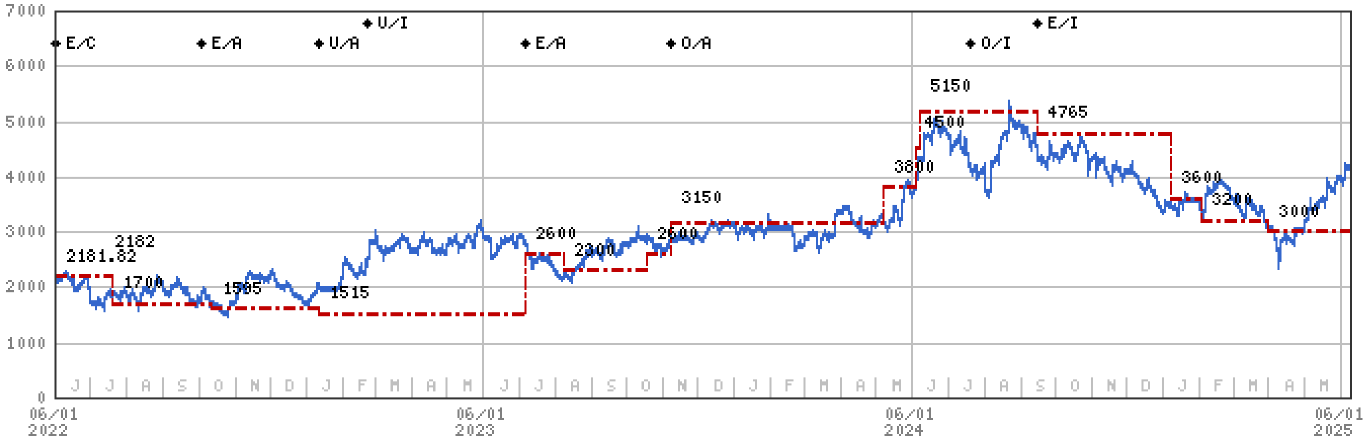
Price Target History: 4/30/20 : 81.5; 6/2/20 : 85; 7/13/20 : 104.75; 7/30/20 : 107.75; 8/23/20 : 130; 10/13/20 : 136; 12/15/20 : 144; 1/20/21 : 152; 1/27/21 : 164; 4/5/21 : 156; 4/15/21 : 157; 4/20/21 : 158; 4/28/21 : 161; 6/23/21 : 162; 7/14/21 : 166; 7/27/21 : 168; 10/26/21 : 166; 10/28/21 : 164; 12/7/21 : 200; 1/27/22 : 210; 4/28/22 : 195; 6/14/22 : 185; 7/19/22 : 180; 10/17/22 : 177; 11/7/22 : 175; 3/3/23 : 180; 5/4/23 : 185; 6/1/23 : 190; 7/16/23 : 220; 8/3/23 : 215; 10/15/23 : 210; 12/7/23 : 220; 4/22/24 : 210; 5/3/24 : 216; 7/14/24 : 273; 1/31/25 : 275; 3/12/25 : 252; 4/8/25 : 220; 4/28/25 : 235

Source: Morgan Stanley Research   Date Format : MM/DD/YY   Price Target ---   No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) ---   Stock Price (Covered by Current Analyst) ---
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Aspeed Technology (5274.TWO) - As of 06/09/25 GMT in TWD
Industry : Greater China Technology Semiconductors



Stock Rating History: 6/1/20 : O/I; 10/12/21 : O/C; 5/20/22 : E/C; 10/4/22 : E/A; 1/11/23 : U/A; 2/22/23 : U/I; 7/7/23 : E/A; 11/8/23 : O/A; 7/21/24 : O/I; 9/15/24 : E/I

Price Target History: 5/25/20 : 1509.09; 8/3/20 : 1309.09; 1/5/21 : 1818.18; 3/24/21 : 1781.82; 4/23/21 : 2036.36; 6/3/21 : 2322.73; 11/16/21 : 3090.91; 4/8/22 : 3636.36; 5/20/22 : 2181.82; 7/13/22 : 2182; 7/19/22 : 1700; 10/12/22 : 1595; 1/11/23 : 1515; 7/7/23 : 2600; 8/8/23 : 2300; 10/19/23 : 2600; 11/8/23 : 3150; 5/7/24 : 3800; 6/3/24 : 4500; 6/7/24 : 5150; 9/15/24 : 4765; 1/8/25 : 3600; 2/3/25 : 3200; 4/1/25 : 3000

Source: Morgan Stanley Research   Date Format : MM/DD/YY   Price Target   No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst)   Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

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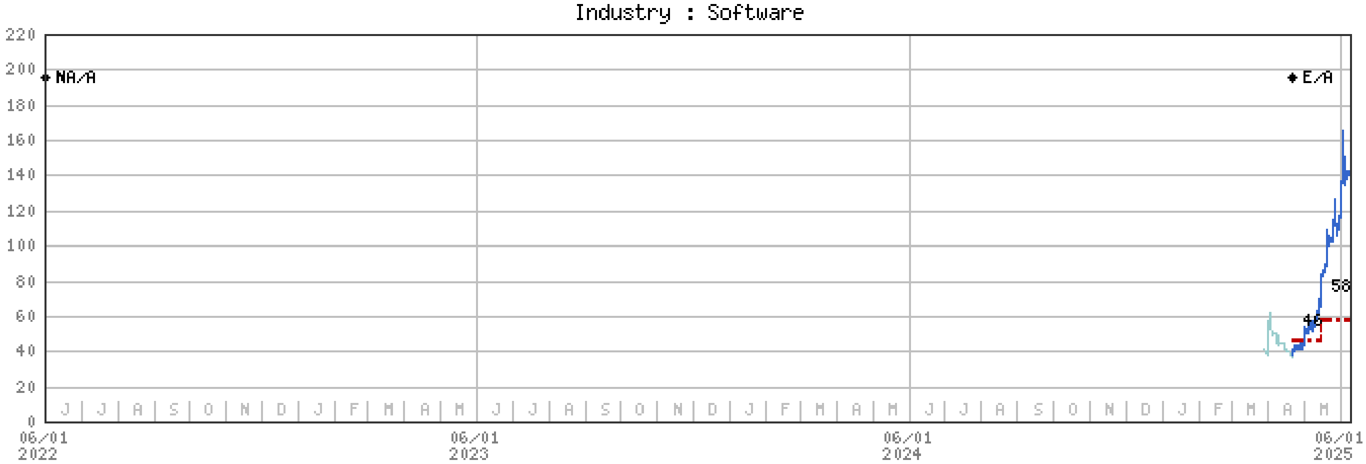
CoreWeave (CRWV.O) - As of 06/09/25 GMT in USD
Industry : Software



Stock Rating History: 6/1/20 : NA/A; 7/27/20 : NA/I; 3/8/21 : NA/A; 4/22/25 : E/A

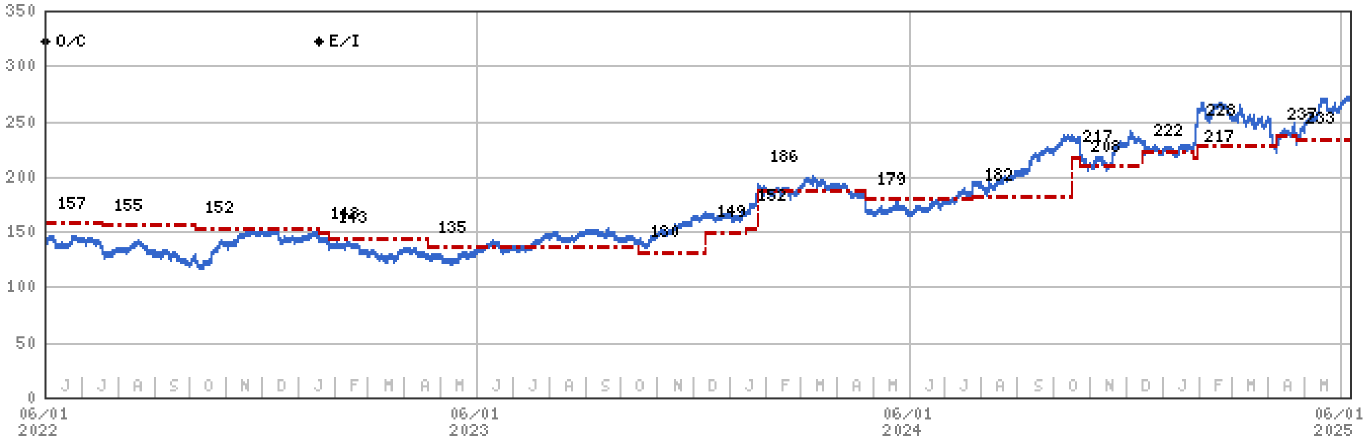
Price Target History: 4/22/25 : 46; 5/15/25 : 58

Source: Morgan Stanley Research   Date Format : MM/DD/YY   Price Target   No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst)   Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

IBM (IBM.N) - As of 06/09/25 GMT in USD
Industry : IT Hardware



Stock Rating History: 6/1/20 : E/I; 10/12/20 : E/A; 6/10/21 : E/I; 10/5/21 : E/C; 4/13/22 : O/C; 1/18/23 : E/I

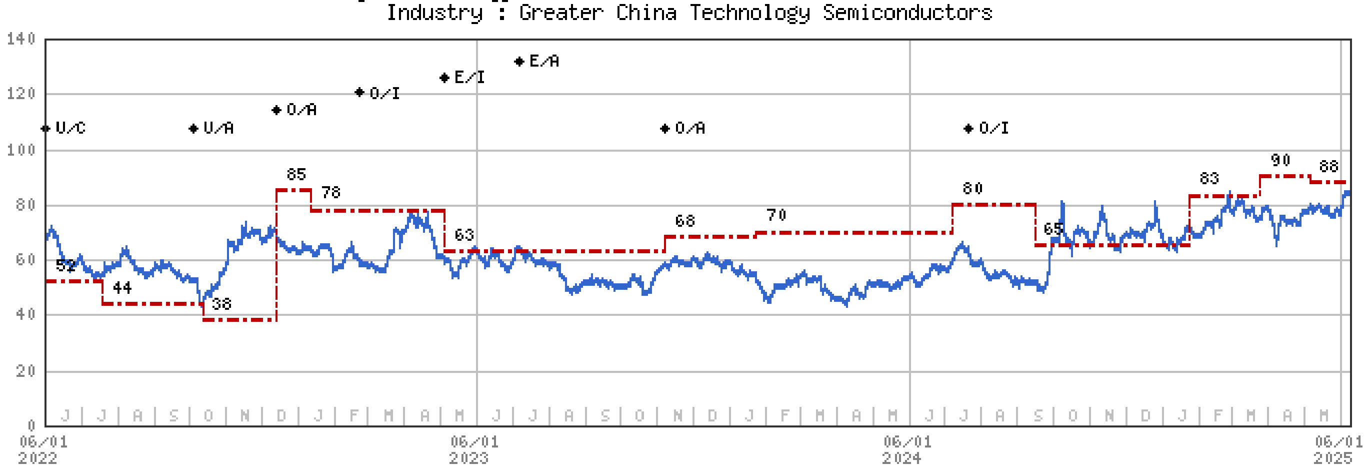
Price Target History: 4/21/20 : 111; 7/20/20 : 128; 10/8/20 : 140; 4/7/21 : 150; 4/19/21 : 152; 7/20/21 : 164; 11/4/21 : 147; 4/13/22 : 150; 4/20/22 : 157; 7/19/22 : 155; 10/5/22 : 152; 1/18/23 : 148; 1/26/23 : 143; 4/20/23 : 135; 10/16/23 : 130; 12/12/23 : 149; 1/15/24 : 152; 1/25/24 : 186; 4/25/24 : 179; 7/25/24 : 182; 10/17/24 : 217; 10/24/24 : 208; 12/16/24 : 222; 1/28/25 : 217; 1/30/25 : 228; 4/8/25 : 237; 4/24/25 : 233

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Montage Technology Co Ltd (688008.SS) - As of 06/09/25 GMT in CNY
Industry : Greater China Technology Semiconductors



Stock Rating History: 6/1/20 : O/I; 5/5/21 : E/I; 10/12/21 : E/C; 5/20/22 : U/C; 10/4/22 : U/A; 12/13/22 : O/A; 2/22/23 : O/I; 5/4/23 : E/I; 7/7/23 : E/A; 11/8/23 : O/A; 7/21/24 : O/I

Price Target History: 4/29/20 : 115; 11/3/20 : 101; 3/24/21 : 88; 5/5/21 : 55; 10/29/21 : 62; 11/16/21 : 80; 4/8/22 : 60; 5/20/22 : 52; 7/19/22 : 44; 10/12/22 : 38; 12/13/22 : 85; 1/11/23 : 78; 5/4/23 : 63; 11/8/23 : 68; 1/23/24 : 70; 7/8/24 : 80; 9/15/24 : 65; 1/24/25 : 83; 3/25/25 : 90; 5/6/25 : 88

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (06/09/2025)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US$24.70
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb180.01
Advanced Wireless Semiconductor Co (8086.TWO)	E (10/24/2024)	NT$76.20

Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT$2,725.00
Andes Technology Corp (6533.TW)	O (08/04/2022)	NT$271.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT$140.00
Global Unichip Corp (3443.TW)	O (07/27/2024)	NT$1,125.00
GlobalWafers Co Ltd (6488.TWO)	E (02/22/2023)	NT$301.50
Gudeng Precision (3680.TWO)	O (08/27/2024)	NT$367.00
Hua Hong Semiconductor Ltd (1347.HK)	E (05/09/2025)	HK$31.70
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT$102.00
M31 Technology Corp (6643.TWO)	E (09/23/2024)	NT$515.00
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb71.17
MediaTek (2454.TW)	O (09/12/2023)	NT$1,285.00
Nanya Technology Corp. (2408.TW)	E (05/15/2025)	NT$51.70
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb430.49
Phison Electronics Corp (8299.TWO)	O (03/11/2025)	NT$535.00
SG Micro Corp. (300661.SZ)	O (06/17/2024)	Rmb97.37
Silergy Corp. (6415.TW)	O (02/13/2025)	NT$378.50
SMIC (0981.HK)	E (04/09/2025)	HK$40.20
TSMC (2330.TW)	O (02/07/2022)	NT$1,005.00
UMC (2303.TW)	E (10/28/2024)	NT$45.65
Vanguard International Semiconductor (5347.TWO)	U (08/05/2024)	NT$89.00
Will Semiconductor Co Ltd Shanghai (603501.SS)	O (05/09/2025)	Rmb132.40
WIN Semiconductors Corp (3105.TWO)	E (05/12/2025)	NT$84.00
Daisy Dai, CFA		
ASMPT Ltd (0522.HK)	E (11/20/2024)	HK$55.25
China Resources Microelectronics Limited (688396.SS)	U (09/02/2024)	Rmb47.31
Elan Microelectronics Corp (2458.TW)	E (04/08/2025)	NT$126.50
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb122.91
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	E (04/09/2024)	Rmb24.26
JCET Group Co Ltd (600584.SS)	U (09/25/2024)	Rmb32.94
Shanghai Anlogic Infotech Co Ltd (688107.SS)	E (05/14/2024)	Rmb27.06
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK$30.70
SICC Co Ltd (688234.SS)	O (09/18/2023)	Rmb58.09
StarPower Semiconductor Ltd (603290.SS)	O (03/01/2022)	Rmb81.05
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb64.45
Universal Scientific Ind. (Shanghai) (601231.SS)	E (10/23/2024)	Rmb14.65
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb49.33
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	E (09/15/2024)	NT$311.50
ASMedia Technology Inc (5269.TW)	E (04/08/2025)	NT$1,980.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT$4,125.00
Egis Technology Inc (6462.TWO)	E (03/26/2024)	NT$103.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb143.57
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb123.44
Macronix International Co Ltd (2337.TW)	U (10/19/2021)	NT$21.85
Montage Technology Co Ltd (688008.SS)	O (11/08/2023)	Rmb83.84
Novatek (3034.TW)	O (03/18/2024)	NT$497.00
Nuvoton Technology Corporation (4919.TW)	O (10/29/2024)	NT$72.10
Parade Technologies Ltd (4966.TWO)	U (06/04/2024)	NT$568.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	E (06/13/2024)	NT$15.65
Realtek Semiconductor (2379.TW)	U (09/15/2024)	NT$550.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	E (06/12/2024)	Rmb69.49
Sino Wealth Electronic (300327.SZ)	U (03/31/2025)	Rmb21.41
Winbond Electronics Corp (2344.TW)	O (03/11/2025)	NT$18.40
WPG Holdings (3702.TW)	E (11/16/2023)	NT$69.70
Duan Liu		

Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb31.35
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	O (03/11/2025)	Rmb73.47
Tiffany Yeh		
AllRing Tech Co. (6187.TWO)	E (05/13/2025)	NT$332.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT$241.00
Himax Technologies Inc (HIMX.O)	O (05/09/2025)	US$8.69
Silicon Motion (SIMO.O)	O (05/06/2024)	US$67.08

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* Historical prices are not split adjusted.